Machine basics, DC machines, rotating fields

Cheat sheet · Day 07 · Fri 30 Oct · pack M6

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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CH4-6 MACHINE BASICS / DC / ROTATING FIELDS
n_S = 60 f / p   (p = POLE-PAIRS; poles = 2p)   = 120 f / poles
  50 Hz: 2p 3000 | 4p 1500 | 6p 1000 | 8p 750 | 10p 600 | 12p 500
Poles from rated speed: p = 60f/n_R -> integer => synchronous;
  non-integer => induction, take p = floor(60f/n_R)
Ideal eff: IM eta = n_R/n_S = 1-s ; DC motor eta = E/V = n/n_NL ; DC gen eta = V/E
Linear τ-n: n = n0 - x (n0 - n_R),  x = τ/τ_R ; n0 = n_NL (DC) or n_S (IM)
  Sync: n = n_S always (vertical line)
  P/P_R = x * n/n_R   (sync: = x)
Speed control: DC -> vary V (speed DROP constant, curve slides parallel)
  AC -> vary f AND V, V/f const (slip speed constant). Target speed at torque x:
  n0_new = n_target + x*dn_R, then f = f_R*n0_new/n0, V = V_R*f/f_R
DC: E = k w, τ = k I, V = E + I R (motor, I into machine); w in rad/s = 2 pi n/60
  k [V/(rad/s) = Nm/A]: no-load k = V/w_NL ; stall R = V/I_S, k = τ_S/I_S
  Pin = VI, Pout = EI = τ w, Ploss = I^2 R
  stall: E=0, I_S=V/R, τ_S=k I_S ; max power at n_NL/2, eta = 50%
  motoring n<n_NL, E<V, I>0 ; generating n>n_NL, E>V, I<0 (E = V + |I|R)
  wind gen + battery: diode stops motoring when E < V
Rotating field: 3 phases 120 deg apart; 6 step directions; balanced 3ph -> constant
  magnitude, constant speed n_S. Swap ANY TWO leads to reverse (cyclic relabel doesn
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